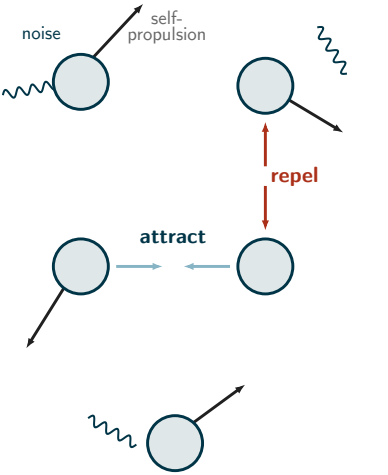


Computational modeling families for collective cell dynamics

Non-exclusive families: agent-based implementations may be particle- or lattice-based; hybrids couple representations.

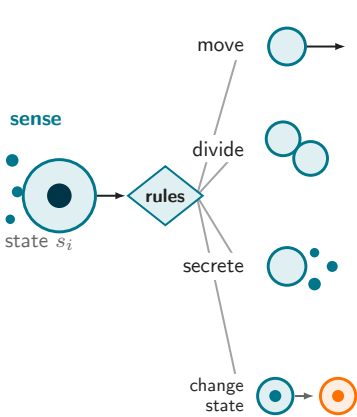
conceptual schematic · not to scale

A Brownian / active particle models



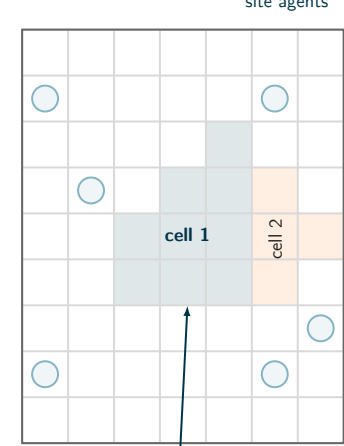
off-lattice positions · stochastic motion · pair forces

B Agent-based models



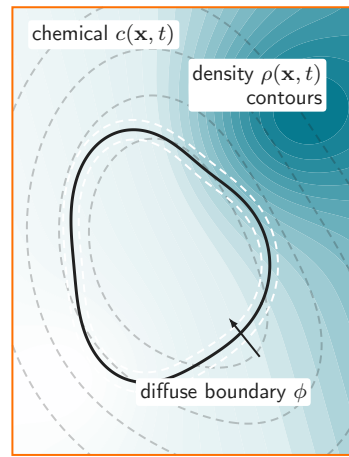
cell-level state + explicit decision rules

C Lattice-based cell models



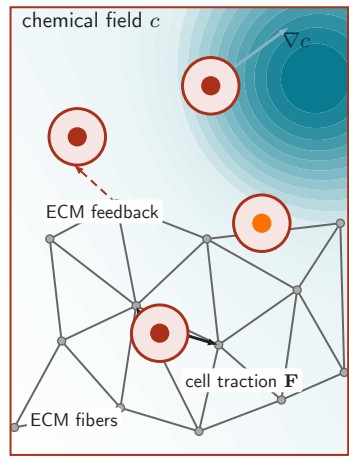
one cell occupies many sites (CPM)

D Continuum / field-based models



smooth state variables defined over space

E Hybrid / multiscale models



discrete cells ↔ fields ↔ matrix mechanics

REPRESENTATION SCALE / RESOLUTION — CROSS-CUTTING, NOT MONOTONIC



Subcellular states and deformability can be embedded across particle, agent-based, lattice, field, and hybrid families; neither defines a separate family.